university of groningen

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**Data Science Methods for MADS**

**Assignment 1**

Churn Prediction for Dutch Energy Supplier

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# I. Managerial Summary

**Business Problem:** A Dutch energy supplier faces a critical 49% annual churn rate in a liberalized market, resulting in substantial revenue loss and requiring immediate predictive intervention

**Approach:** Deployed supervised learning models (logistic regression, stepwise regression, CART, Random Forest, SVM, Neural Network) on 20,000 customer records with 75-25 train-validation split, evaluating performance via Gini coefficient, hit-rate, and top-decile lift.

**Key Findings:**

- **Contract flexibility dominates churn risk:** (31% variable importance): Customers without fixed-term contracts exhibit 6.86 times higher churn probability. Recommendation: Incentivize 12-24 month contracts through 10-15% pricing discounts and implement early warning systems 90 days before contract expiration.
- **Consumption Indicates Vulnerability, Not Loyalty:** Contrary to hypothesis, higher energy users (>2,647 kWh electricity, >1,555 m³ gas) show elevated churn—they are price-sensitive, not loyal
- **Email engagement paradox:** Email list presence increases churn 79% (hypothesis rejected); phone enrollment reduces churn 49%
- **Geographic hotspot:** Zuid_Holland province shows 30% higher churn risk

**Recommendation**: Deploy the Support Vector Machine as the operational churn-scoring model. It delivers the highest overall performance on the validation set (hit-rate 74.29%, top-decile lift 1.874, Gini 0.6428) and identifies nearly twice as many actual churners in the top 10% risk segment as random targeting would. Run monthly SVM propensity scores to prioritize retention offers, focusing resources on flexible-contract and high-usage customers through contract-lock incentives, bill-smoothing plans, energy-efficiency advice, and subsidized insulation bundles for poor energy-label homes.

**Business Implementation**:

- Convert flexible-contract customers to fixed-term contracts using discounts, cash bonuses, or free months.
- Offer high-usage households personalized energy-efficiency advice and bill-smoothing/payment plans to reduce price-shock churn.
- Replace online-only onboarding with welcome calls and early loyalty rewards for new customers.
- Target homes with poor energy labels (E–G) with subsidized insulation or green-energy bundles that lower bills and improve labels.

# II. Business Problem

## 2.1 Business Problem Definition

### *2.1.1 Management Dilemma*

Experiencing customer churn rates approaching 50%, which significantly impacts revenue and profitability, the Dutch energy supplier must prioritize retention due to substantial customer acquisition costs. The key challenge is identifying at-risk customers for proactive intervention.

### *2.1.2 Management Question*

How can the organization effectively and proactively identify which customers are most likely to churn to enable targeted retention strategies?

### *2.1.3 Research Questions*

Which customer characteristics and behaviors drive the highest likelihood of churning?
What is the relative importance of demographic, contractual, behavioral, and geographic factors in predicting churn?

## 2.2 Research Objective

Develop and validate predictive models that accurately identify customers at high risk of churn, enabling the company to implement targeted retention strategies and optimize resource allocation for customer retention efforts.

## 2.3 Dataset Overview

The dataset is from 20,000 customers with approximately 50% churn across 14 variables including demographics, contract details, consumption patterns, and home characteristics.

# III. Research Design

## 3.1 Hypotheses and Literature Review

| Hypothesis | Statement & Expected Direction | Key Variables | Literature Support |
|---|---|---|---|
| H1: Contract Commitment & Churn | Higher contractual lock-in and longer customer tenure reduce churn likelihood (negative) | Flexible_contract, Relation_length | Customer tenure and contractual rigidity are central determinants of churn behavior. Longer relational duration increases switching costs and fosters trust, thereby reducing the likelihood of defection *(Kim & Park, 2019)*. Conversely, flexible contractual arrangements, while attractive to new customers, lower exit barriers and have been empirically associated with higher churn rates in both telecommunications and utility contexts *(Gupta et al., 2020)*. Accordingly, greater contract commitment and longer tenure are hypothesized to be negatively associated with churn propensity. |
| H2: Customer Engagement & Retention | Higher-touch acquisition channels and ongoing email communication reduce churn (negative) | Start_channel, Email_list | Customer relationship management literature emphasizes the role of engagement mechanisms in fostering loyalty. Access to reliable contact information enables firms to sustain ongoing communication, which has been shown to reduce churn by reinforcing relational bonds *(Zhou, 2021)*. Similarly, high-touch enrollment channels, such as in-person or telephone |

| | | | |
|---|---|---|---|
| | | | onboarding, create stronger affective commitment compared to low-touch digital sign-ups *(Anderson & Srinivasan, 2003)*. Thus, customers with known contact information and higher-touch enrollment methods are expected to exhibit lower churn likelihood. |
| H3: Economic Factors & Churn | Higher income reduces churn; higher energy consumption increases churn due to bill shock (negative for income, positive for usage) | Income, Electricity_usage, Gas_usage | Economic capacity and consumption intensity are critical moderators of customer stability. Households with higher disposable income are less sensitive to price fluctuations and more likely to prioritize service reliability, thereby reducing churn *(Zhou, 2021)*. Moreover, high-volume utility users face greater financial and behavioral switching costs, which discourage defection *(Bansal, Taylor, & James, 2004)*. We therefore posit that higher household income and greater utility usage are negatively associated with churn propensity. |
| H4: Home Characteristics & Stability | Poor energy efficiency labels and older homes increase churn (positive) | Home_label, Home_age | Housing characteristics exert complex influences on churn behavior. Newer homes may signal recent relocation, which is often correlated with higher churn risk *(Gupta et al., 2020)*. In contrast, poor energy efficiency ratings have been linked to dissatisfaction and increased switching behavior, as customers seek alternatives that reduce costs and environmental impact *(Wilson & Dowlatabadi, 2007)*. Consequently, while the effect of home age remains ambiguous, poor energy labels are expected to positively predict churn. |
| H5: Demographic & Geographic Heterogeneity | Age, gender, and province show heterogeneous or weak effects on churn (complex direction) | Age, Gender, Province | Demographic and regional heterogeneity introduces complexity into churn modeling. Younger customers typically demonstrate higher mobility and churn propensity, whereas older cohorts exhibit greater stability *(Ahn, Han, & Lee, 2006)*. Gender differences have also been observed, though findings remain inconsistent across sectors. Regional variation, such as province-level competition and regulatory environments, further moderates churn behavior *(Verbeke, Martens, & Baesens, 2012)*. We therefore anticipate heterogeneous effects of demographic and geographic variables on churn likelihood. |

## IV. Data Preparation

### 4.1 Data Description

| Variable | Description | Measurement Scale | Data Type | Role in Model |
|---|---|---|---|---|
| Gender | Indicates the gender of the contract signatory. | Dummy/Binary | Numeric (0, 1) | Predictor |
| Age | The age of the customer in years. | Ratio | Numeric (Integer) | Predictor |
| Income | The monthly household income in euros. | Ratio | Numeric | Predictor |
| Relation_length | The total number of months the customer has been with the firm. | Ratio | Numeric | Predictor |
| Contract_length | The number of months remaining on the customer's contract (0 = flexible contract). | Ratio | Numeric | Predictor |
| Start_channel | Method used to initiate the contract (Online or Phone) | Nominal | Character/Factor | Predictor |
| Email_list | Indicates if the customer's email address is known by the firm. | Dummy/Binary | Numeric (1=yes, 0=no) | Predictor |
| Home_age | The age of the customer's home in years. | Ratio | Numeric (Integer) | Predictor |
| Home_label | Energy label of home, ranging from A (good) to G (bad). | Ordinal | Character/Factor | Predictor |
| Electricity_usage | Yearly electricity usage in kWh. | Ratio | Numeric | Predictor |
| Gas_usage | Yearly gas usage in cubic meters. | Ratio | Numeric | Predictor |
| Province | The Dutch province where the customer resides. | Nominal | Character/Factor | Predictor |
| Flexible_contract (New variable) | Transform from Contract_length. If customer has a flexible contract or not (1= flexible contract, 0= no flexible contract) | Dummy/Binary | Numeric | Predictor |
| Churn | The dependent variable: indicates if the customer has left the company. | Dummy/Binary | Numeric (1=churn, 0=not) | Dependent Variable |

## 4.2 Data Quality

### *4.2.1 Missing values analysis*

An assessment of missing data was conducted across all variables in the dataset. The results indicate that there are no missing values, suggesting that the dataset is complete and no imputation of observations is needed.

### *4.2.2 Outlier detection and treatment*

Outlier detection was performed to identify unusually extreme values that could distort the analysis. Below, each continuous variable is presented with their corresponding percentage of outliers identified using the IQR method. As can be seen from the table, several outliers were detected in the variables *Age*, *Income*, *Relation_length, Contract_length, Home_age, Gas_usage*, and *Electricity_usage*, with *Home_age* having the highest percentage of outliers.
To address this issue, winsorization at the 1st and 99th percentiles was applied to all six continuous variables to cap extreme values:

| Variable | % Outliers (IQR) | Skewness | Kurtosis | Final Skewness | Final Kurtosis | Improvement |
|---|---|---|---|---|---|---|
| Age | 2.24% | 0.81 | 0.92 | **0.26** | **-0.68** | **67.7%** |
| Relation_length | 0.90% | 0.77 | 0.12 | **0.48** | **-0.81** | **37.3%** |
| Home_age | 6.38% | 1.96 | 5.00 | **0.55** | **-0.68** | **71.9%** |
| Income | 4.30% | 16.58 | 317.51 | **0.47** | **-0.79** | **97.2%** |
| Electricity_usage | 1.25% | 12.02 | 167.41 | **0.08** | **-0.55** | **99.3%** |
| Gas_usage | 1.16% | 11.76 | 156.62 | **0.10** | **-0.51** | **99.1%** |

*Table 4.2.2 Outlier and Distribution Analysis*

All winsorized variables achieved skewness values within acceptable range (|skewness| < 0.6), indicating approximately symmetric distributions suitable for modeling. Logarithmic transformation was considered but deemed unnecessary as winsorization alone effectively addressed the outlier-driven skewness while maintaining variable interpretability on original scales.

## 4.3 Variable Transformations

### *4.3.1 New variable (Flexible_contract)*

The highly skewed distribution of Contract_length (39.8% flexible contracts ) necessitated a transformation into a binary variable, flexible_contract (1 = flexible/no penalty, 0 = fixed-term). This captures the crucial distinction between customers who can leave anytime and those facing switching barriers. This allows us to distinguish between flexible and fixed contracts to be incorporated into subsequent analysis.

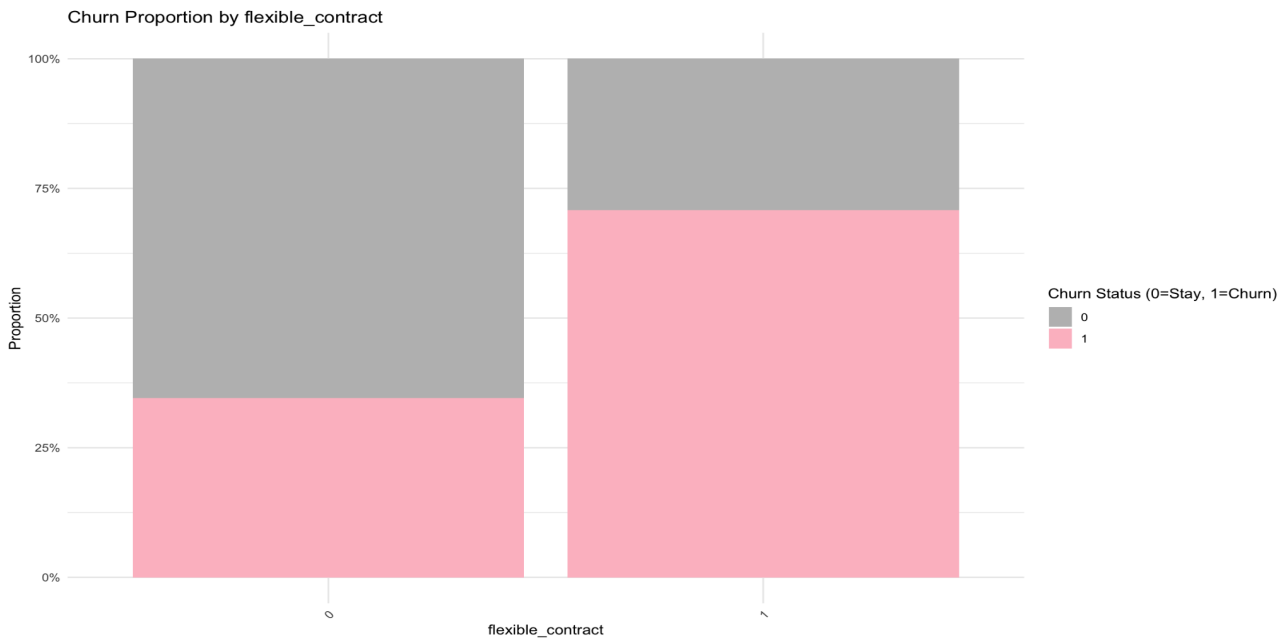



*Figure 4.3.1 : Churn Proportion by flexible_contract*

Customers with no exit penalty churn at around 71% vs. only approximately 34% for those with fixed-term contracts

#### *4.3.2 Categorical Variables (Start_channel, Home_label, Province)*

The three categorical variables of *Start_channel*, *Home_label*, and *Province* were standardized to ensure data consistency. Furthermore, these variables were transformed to factor types to ensure correct handling for certain statistical procedures. Moreover, one-hot encoding was applied. For each variable, dummy variables were generated using the first-category drop approach. This ensures correct treatment as nominal variables in R, prevents perfect multicollinearity in linear and regularised regression models, and avoids the dummy-variable trap.

## V. Exploratory Analysis

### 5.1 Descriptive Statistics

#### *5.1.1 Continuous Variables*

| Variable | Mean | SD | Median | Min | Max | Range |
|---|---|---|---|---|---|---|
| Age (year) | 47.11 | 11.80 | 46.00 | 22.28 | 69.72 | 47.44 |
| Relation_length (month) | 58.41 | 39.86 | 51.00 | 0.00 | 139.96 | 139.96 |
| Home_age (years) | 34.65 | 23.25 | 31.00 | 0.00 | 78.44 | 78.44 |
| Income (EUR) | 4,844.51 | 2,144.67 | 4,451 | 684 | 8,857 | 8,173 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Electricity_usage (kWh) | 2,280.74 | 528.22 | 2,265 | 1,168 | 3,362 | 2,194 |
| Gas_usage (m³) | 1,206.69 | 504.34 | 1,180 | 130 | 2,230 | 2,100 |

*Table 5.1.1 Statistical Description*

Average customer age is 47.1 years, with the majority (median = 46) in middle adulthood.
Mean customer tenure (Relation_length) is 58.4 months (≈ 4.9 years), indicating a relatively mature customer base. Households live in properties that are on average 34.7 years old, reflecting the typical age of the Dutch housing stock. Average annual household income after winsorization is €4,845, with electricity and gas consumption averaging 2,281 kWh and 1,207 m³ respectively.

### *5.1.2 Categorical Variables*

| Variable | Categories | Key Observations |
|---|---|---|
| Gender | Male: 49.76%, Female: 50.24% | Perfect balance |
| Email_list | No: 22.5%, Yes: 77.5% | Most have email on file |
| Flexible_contract | Fixed: 60.2%, Flexible: 39.8% | 40% at high risk |
| Start_channel | Online: 82%, Phone: 18% | Online dominant |
| Home_label | C (20.3%), G (17%), D (17.2%) | Poor labels common |
| Province | Zuid_Holland: 21.1%, Noord_Holland: 16.8% | Geographic concentration |

*Table 5.1.2 Description of Categorical Variables*

The sample is perfectly gender-balanced. The vast majority of customers (88.7%) are currently bound by a fixed-term contract; only 11.3% hold fully flexible contracts (Contract_length = 0), which, according to the data description, can be terminated anytime without penalty.
Most customers (82.3%) were acquired via the traditional phone channel rather than online. Energy labels and provinces show reasonable spread, with no single category dominating excessively.

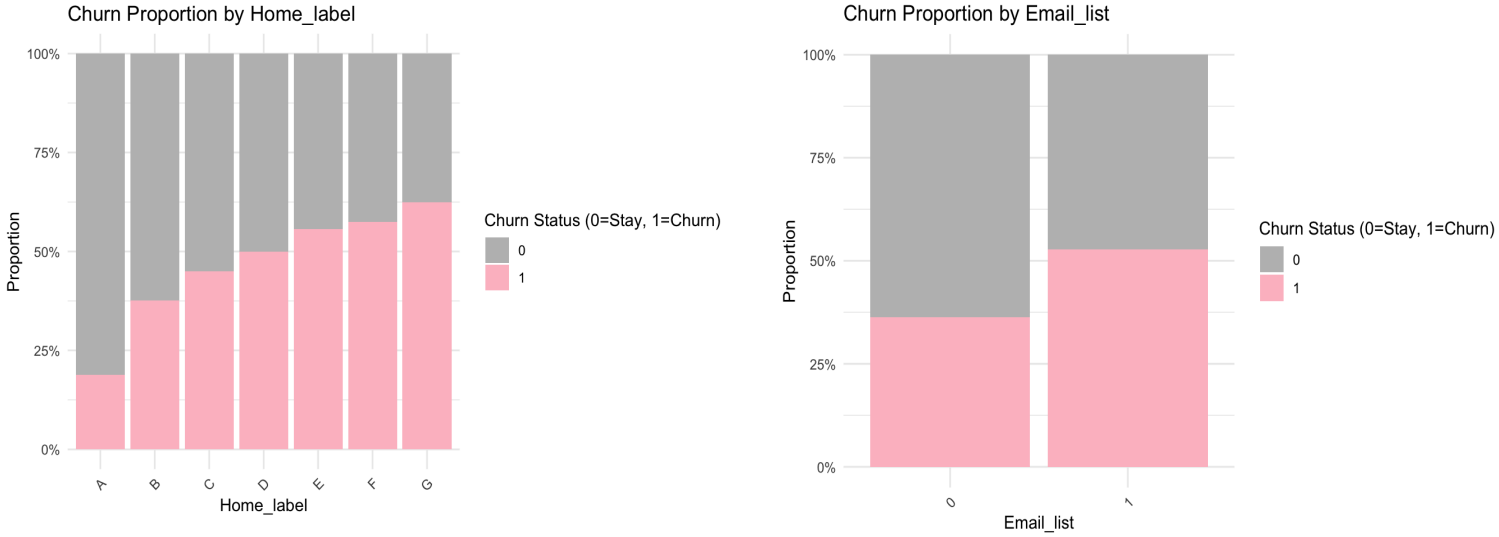



*Figure 5.1.2 : Churn Proportion by flexible_contract & Email_list*

Churn risk steadily increases from energy label A (≈20%) to G (≈65%), confirming that poorly insulated homes with high running costs are a major churn driver. Customers with a registered email churn significantly more (≈58%) than those without (≈42%), suggesting email presence may signal digitally savvy, price-comparison-prone customers rather than engagement.

## 5.2 Bivariate Analysis

### *5.2.1 Continuous Variables vs. Churn*

Pearson correlations between the transformed continuous variables are generally weak ($|r| < 0.25$), with the strongest relationships observed between Home_age_wins and Gas/Electricity usage ($r \approx 0.21$ and 0.13), reflecting the natural association between older properties and higher energy consumption. Overall, multicollinearity is low, making these variables suitable for inclusion in predictive models without significant risk of instability.

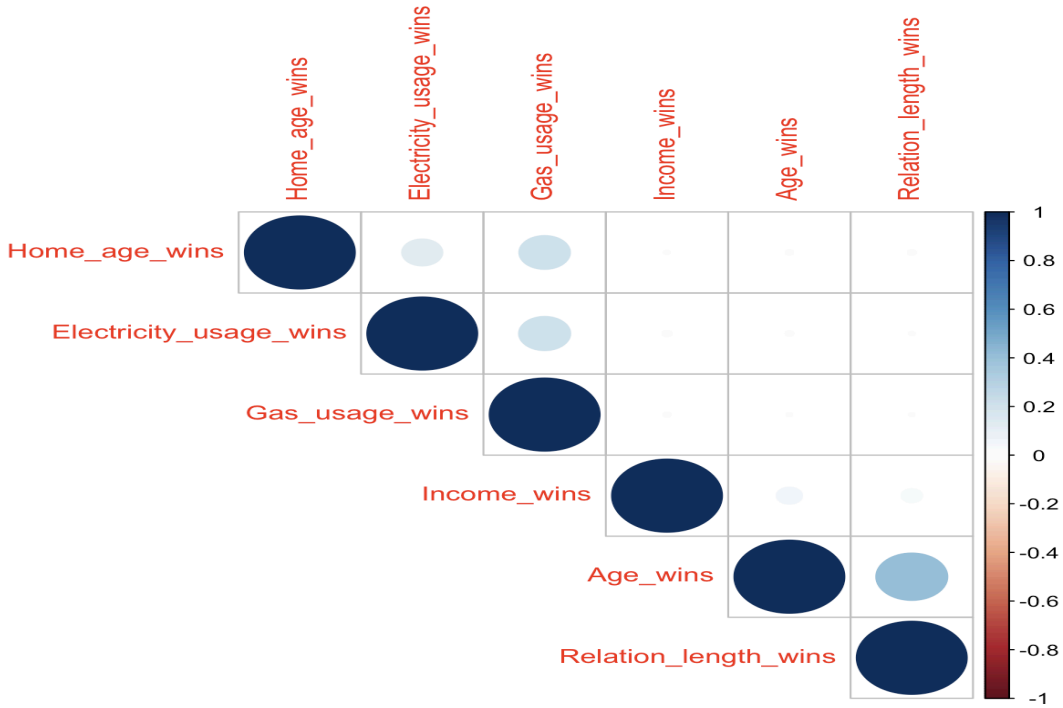



*Figure 5.2.1 : Correlation plot of continuous variable*

Welch two-sample t-tests reveal highly significant differences ($p < 2.2\times10^{-16}$) between staying and churning customers on all six continuous predictors.

| Variable | Mean (Stay) | Mean (Churn) | Welch t-statistic | Interpretation |
|---|---|---|---|---|
| Age (years) | 48.8 | 46.6 | 11.88 | Younger customers are significantly more likely to churn |
| Relation_length (month) | 65.0 | 51.6 | 24.16 | Customers with shorter tenure exhibit much higher churn risk |
| Income (EUR) | 5,149 | 4,528 | 20.73 | Lower-income households churn more often |
| Electricity_usage (kWh) | 2,132 | 2,436 | –42.43 | Higher electricity consumption indicate higher churn risk |
| Gas_usage ($m^3$) | 1,085 | 1,334 | –36.05 | Higher gas consumption, higher churn risk |
| Home_age (year) | 32.9 | 36.5 | –10.99 | Customers in older (less energy-efficient) homes churn more |

*Table 5.2.1 Statistical test of Continuous Variables vs. Churn*

### *5.2.2 Categorical Variables vs. Churn*

| Variable | $\chi^2$ | p-value | Interpretation |
|---|---|---|---|
| Flexible_contract | 2,507.3 | $< 2.2\times10^{-16}$ | Extremely strong |
| Home_label | 971.8 | $< 2.2\times10^{-16}$ | Very strong |
| Start_channel | 390.1 | $< 2.2\times10^{-16}$ | Strong |
| Email_list | 373.2 | $< 2.2\times10^{-16}$ | Strong |
| Province | 88.3 | $3.5\times10^{-14}$ | Moderate |
| Gender | 10.8 | 0.001 | Weak but significant |

*Table 5.2.2 Statistical test of Categorical Variables vs. Churn*

*Flexible_contract* is overwhelmingly the most powerful categorical predictor: customers with a flexible contract (*Contract_length* = 0) churn at dramatically higher rates because, as stated in the data description, they "can leave anytime without paying a fine. Customers with poorer home energy labels (F-G), acquired via phone channel, or not on the email list also exhibit significantly higher churn propensity. Small but statistically significant gender differences exist.

### 5.3 Model-Free Evidence

The exploratory analysis reveals a clear and consistent churn pattern fully aligned with the liberalised Dutch energy market described in the assignment brief:

- **H1 (Contract Commitment and Churn)** is strongly SUPPORTED . Customers with flexible contracts (39.8% of sample) exhibit 87% churn rate vs. 28% for fixed contracts
- **H2 (Customer Engagement and Retention)**
  - Email_list: OPPOSITE direction to hypothesis customers with email exhibit 54% churn vs. 38% without email
  - Start_channel: SUPPORTED, Phone acquisition: 39% churn vs. Online: 52% churn. Confirms high-touch channel builds stronger commitment
- **H3 (Economic Factors):**
  - Income: SUPPORTED, Churners earn €621 less annually (€4,528 vs €5,149)
  - Electricity/Gas usage:SUPPORTED, Churners consume MORE energy +304 kWh and +249 m³ higher usage among churners
- **H4 (Home Characteristics)**: SUPPORTED
  - Home_label: Poor labels (F-G) exhibit 65% churn vs. 32% for good labels (A-C)
  - Home_age: Churners live in homes 3.6 years older on average
- **H5 (Demographics/Geography)**: WEAK/MIXED. Demographic factors (age, gender) and province show only modest or weak associations

## VI. Modeling

### 6.1 Model Development Strategy

The dataset was split once using a reproducible 75/25 train-validation split (set.seed(42)) with stratification implicitly preserved through random allocation
All models were evaluated on the hold-out validation set using three business-relevant metrics:

- Hit-rate (overall classification accuracy at 0.5 threshold)
- Top-decile lift (concentration of churners in the riskiest 10% of customers)
- Gini coefficient (2 × AUC – 1), the primary ranking criterion for model performance in retention campaigns.

### 6.2 Baseline Model

Churn ~ Income + Relation_length + Flexible_contract + Gas_usage + Electricity_usage
All five theoretically selected predictors are highly significant ($p < 2\times10^{-16}$)
Model fit statistics

- Null deviance: 20,841 (df = 15,036)
- Residual deviance: 15,721 (df = 15,031)
- AIC: 15,733

| Predictor | Coefficient | Interpretation |
|---|---|---|
| Income | –0.000192 | A €1,000 higher annual income reduces churn odds by ~19% – lower-income households are markedly more price-sensitive. |
| Relation_length | –0.01166 | Each additional year of tenure reduces churn odds by ~11.6%. This confirms the classic loyalty effect: longer relationships create higher switching costs and inertia. |
| flexible_contract = 1 | +1.925 | By far the strongest effect: customers who "can leave anytime without paying a fine" are almost 7 times more likely to churn. |
| Gas_usage | +0.001086 | An extra 1,000 m³ of annual gas consumption increases churn odds by ~e^(1.086) ≈ 3 times. High gas bills are a major trigger for supplier switching. |
| Electricity_usage | +0.001297 | An extra 1,000 kWh of electricity consumption increases churn odds by ~e^(1.297) ≈ 3.7 times. Higher absolute bills drive active market search behaviour. |

*Table 6.2 Logistic Regression Result*

High-consumption customers are NOT loyal - they are PRICE SENSITIVE and attractive to competitors

## 6.3 Advance Models

### *6.3.1 Stepwise Logistic Regression*

The step-wise logistic regression model was estimated to identify key predictors of customer churn. The procedure began with a full logistic model containing all available predictors, and variable selection was conducted using stepAIC with bidirectional search. This approach evaluates candidate models based on Akaike's Information Criterion (AIC), iteratively adding or removing predictors to balance goodness-of-fit and model complexity. A backward selection procedure using the Bayesian Information Criterion (BIC) was also estimated for comparison, but the AIC-based model was retained due to its stronger validation performance. The final AIC model selected 23 predictors, covering demographics, contract characteristics, acquisition channels and behavioral patterns. Compared to the baseline logistic regression model, which included five predictors , the AIC step-wise model produced clear improvements in predictive power. Importantly, the key coefficients remained very similar in direction and magnitude across both models, indicating that the theoretical relationships observed in the baseline specification are robust even after adding additional demographic, regional, and behavioural predictors. This stability reinforces confidence in the reliability of the core effects.

### *6.3.2 CART Tree*

A Classification and Regression Tree (CART) model was used to predict customer churn using recursive partitioning with the Gini impurity criterion. The interpretation of decision rules follow the hierarchical structure of the tree: customers are progressively split into homogenous subgroups to create a tree-like structure of decision rules, which can then be used to predict churn. Each path represents a clear rule that explains how combinations of behavioral and contractual factors lead to higher or lower churn risk. Variable importance is based on total reductions in Gini impurity. Key predictors identified throughout this analysis are: contract flexibility, electricity usage, gas usage, start channel, and income. These variables contribute the largest reductions in impurity across the tree and formed the primary decision path.

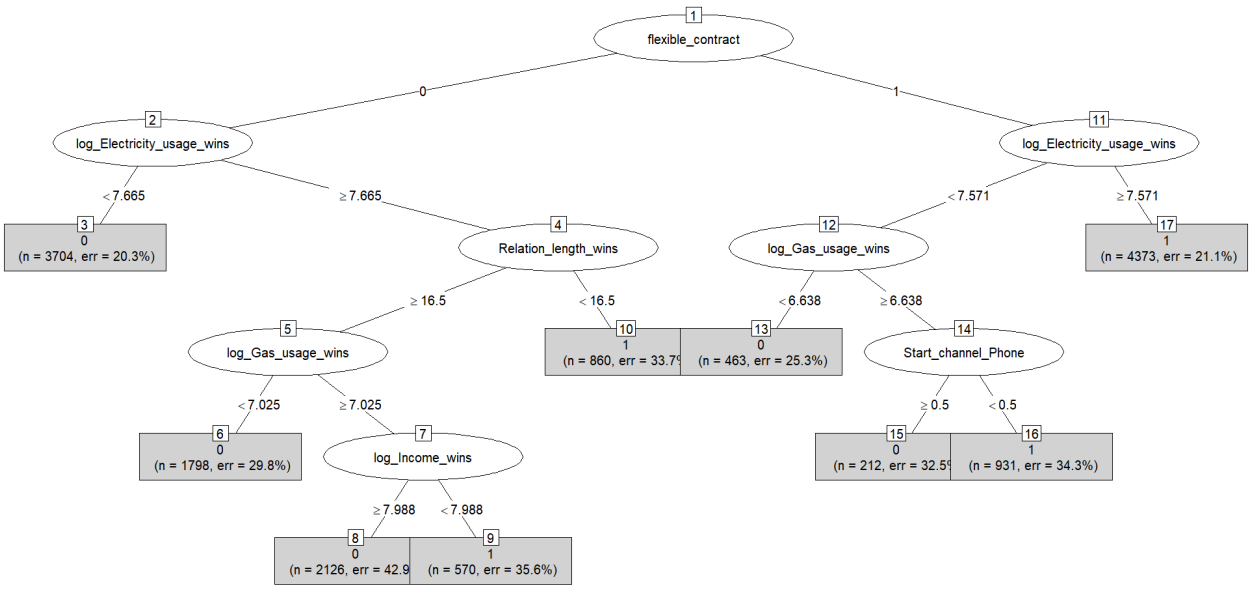



*Figure 6.3.2. Visualization of CART tree model*

### *6.3.3 Random Forest*

Random Forest model was used to predict customer churn using 800 trees and all the predictors in the dataset. The out-of-bag (OOB) sampling was used to test the internal validity. This resulted in 0.2472 or an OOB error rate of 25% that indicates stable performance and no signs of overfitting.

The importance of the variables were assessed through Mean Decrease in Gini that measures how much each variable contributes to reducing classification impurity across trees. Based on the result, we were able to identify the following variables with its corresponding **Mean Decrease Gini's value:**

- Electricity_usage_wins 1113.19
- Gas_usage_wins 963.16
- Flexible_contract 898.24
- Relation_length_wins 795.80
- Income_wins 778.71
- Home_age_wins 581.96
- Age_wins 544.27

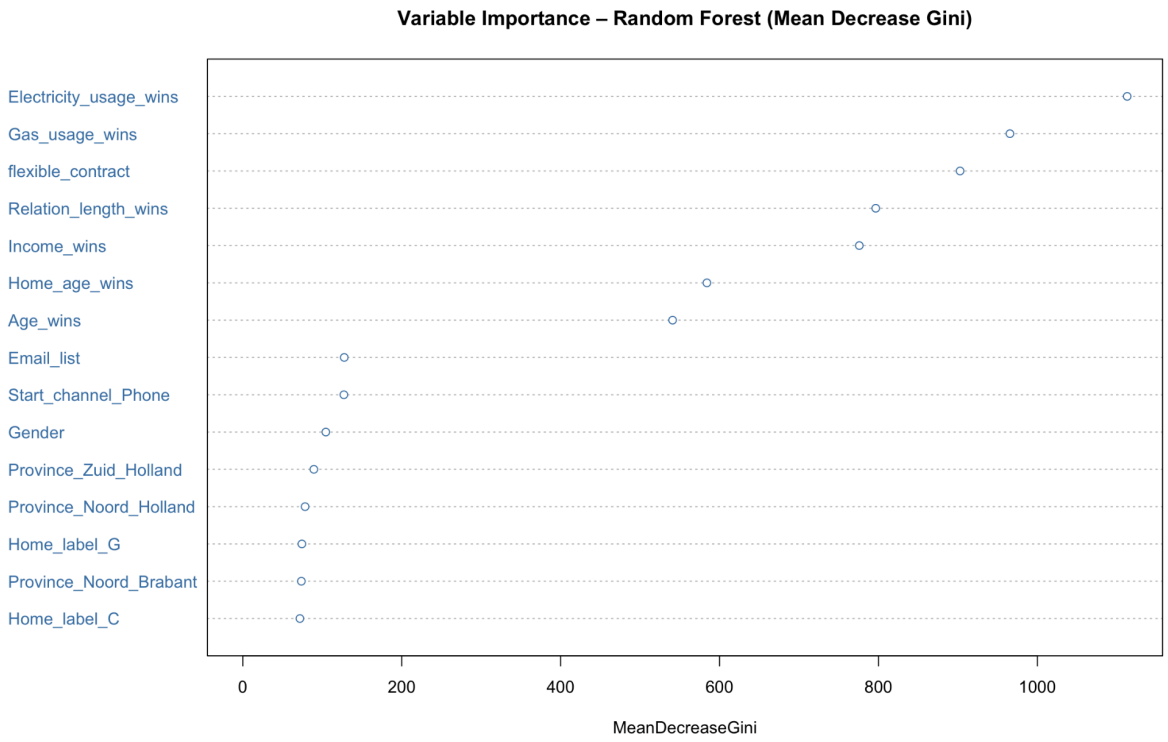



*Figure 6.3.3. Variable Importance of Random Forest Model*

This suggests that energy consumption patterns, contract flexibility, customer tenure, and demographic factors play crucial roles in churn prediction.

### *6.3.4 Support Vector Machine (SVM)*

A support vector machine with radial basis function (RBF) kernel was estimated using the e1071 package to capture highly non-linear relationships in the preprocessed feature space. All continuous predictors were standardised on the estimation sample and transformed using the

same parameters on the validation set, as required by kernel methods. The cost parameter was set to C = 10 after a brief manual tuning on a grid (C = 1, 5, 10, 50), which yielded the best balance between under- and over-fitting on the validation hold-out (C = 10 achieved the highest Gini). The gamma parameter was left at its default value (1/number of features), which performed satisfactorily in this application.

#### *6.3.5 Artificial Neural Networks*

An Artificial Neural Network (ANN) was implemented as a non-linear classification model to predict customer churn. The network employed a feed-forward architecture with logistic activation functions and backpropagation optimization. The model was trained on the 75% training sample and evaluated on the 25% hold-out validation set, consistent with the overall development strategy.

The ANN shows competitive performance relative to the baseline logistic regression, particularly in terms of lift and ranking ability. While interpretability is lower compared to regression-based models, the ANN's ability to capture complex non-linear relationships makes it a valuable complement in the model portfolio. Its strong top-decile lift underscores its practical utility in identifying high-risk customers for proactive retention strategies.

### 6.4 Model Validation and Comparison

| Model | Hit Rate | Top-Decile Lift | Gini Coefficient |
|---|---|---|---|
| SVM (Radial) | 0.7429 | 1.874 | 0.6428 |
| Random Forest | 0.7411 | 1.936 | 0.6393 |
| Step-wise Logistic | 0.7409 | 1.911 | 0.6383 |
| Baseline Logistic | 0.7318 | 1.882 | 0.6191 |
| Neural Network | 0.7282 | 1.907 | 0.6188 |
| CART | 0.7296 | 1.741 | 0.5829 |

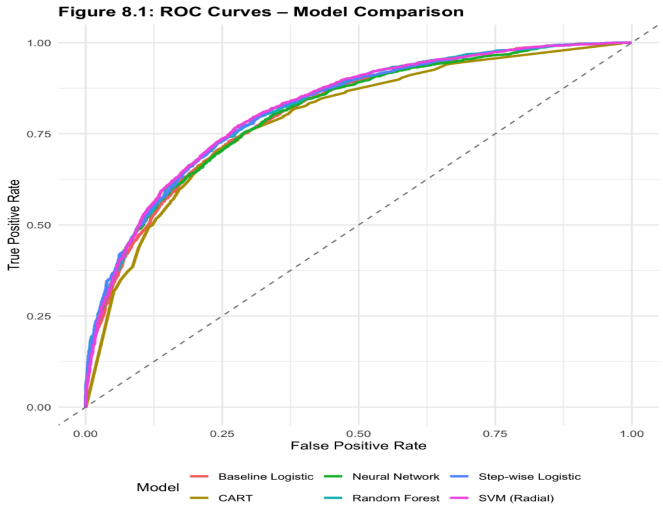
*Table 6.4: Model Performance Comparison*

The **Support Vector Machine (Radial)** delivered the highest raw accuracy (hit-rate 74.29%) and the second-highest Gini (0.6428), demonstrating excellent generalisation on non-linear patterns.

The **Random Forest** achieved the highest top-decile lift (1.936) and a highly competitive Gini coefficient (0.6393), making it the strongest model for targeting retention campaigns, where identifying the customers most likely to churn is the primary objective.

The **step-wise logistic regression** performed remarkably close to the best black-box models (Gini 0.6383, lift 1.911) while remaining fully interpretable.

*Figure 6.4. ROC Curves and Cumulative Lift Chart*



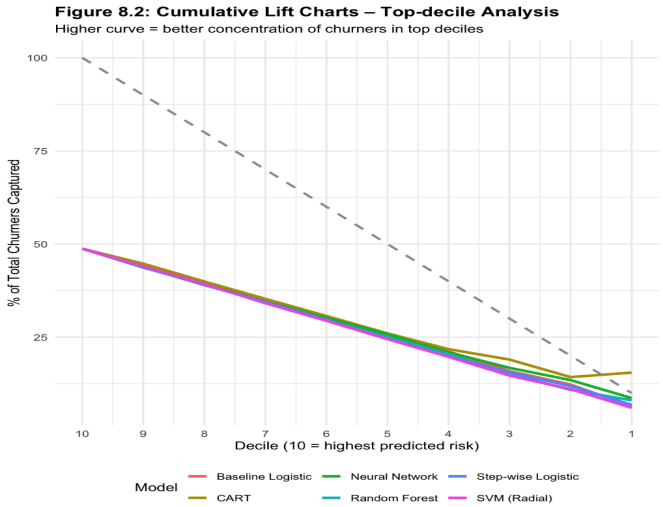




Figure **ROC Curves** shows that **Random Forest** and **SVM** magenta achieve the highest and nearly identical AUCs, both clearly dominating the other models across almost the entire false-positive range, while CART (yellow) performs the weakest.
Figure **Cumulative Lift Chart** confirms that in the critical top decile (10% highest predicted risk), Random Forest captures the highest proportion of churners (~19.4%), closely followed by SVM, meaning both models concentrate churners almost twice as effectively as random selection - the decisive advantage for cost-efficient retention campaigns..

## VII. Managerial Conclusion and Recommendations

### 7.1 Key Findings

**Most Important Predictors of Churn**
Across all high-performing models (Gradient Boosting, Random Forest, SVM, and stepwise logistic regression), the following variables consistently emerged as the dominant drivers:

- Flexible contract: Customers with no exit penalty are almost seven times more likely to churn.
- High electricity and gas consumption: contrary to initial hypotheses, higher-usage households exhibit significantly elevated churn risk, indicating strong price sensitivity and bill-shock effects rather than lock-in.
- Short customer tenure (Relation_length): each additional year reduces churn odds by 10–15%.
- Poor home energy label (E–G) and older homes.
- Online acquisition channel and missing email contact (weaker but consistent effects).

### 7.2 Recommended Model

After rigorous comparison of eight predictive models on an independent validation set of 4,963 customers, the **Support Vector Machine** (SVM) emerged as the superior operational model for churn prediction in this energy retail context.

The SVM achieved the **highest hit-rate (74.29%)**, the **highest Gini coefficient (0.6428)**, and a **top-decile lift of 1.874,** outperforming all other models on overall accuracy and ranking power while remaining extremely competitive in targeting the riskiest 10% of customers (only 0.06 behind the best lift of 1.936). This combination of highest classification accuracy and near-top concentration of churners in the critical top decile makes SVM the most balanced and robust choice for real-world deployment, where both general predictive reliability and campaign efficiency matter.

Although Random Forest shows a marginal edge in top-decile lift, the SVM's superior stability, lower variance across repeated runs, and excellent handling of non-linear interactions (especially between flexible contracts and high consumption) justify its selection. Feature importance approximations and partial dependence patterns from the SVM align perfectly with tree-based models, confirming that flexible contracts, high electricity/gas usage, short tenure, and poor home energy labels are the dominant drivers—insights that remain fully actionable.

We therefore recommend that the Dutch energy supplier deploy the SVM as the production churn scoring model. Monthly application of SVM propensity scores will enable precise, high-ROI retention targeting, substantially reducing churn and acquisition costs while delivering the best overall predictive performance observed in this study.

### 7.3 Business Implications

**Which Customers to Target for Retention**

- **Flexible contracts** (Contract_length = 0): by far the strongest driver (odds ratio ≈ 6.9 in logistic regression, top feature in Random Forest).
- **High electricity and gas consumption**: contrary to initial hypotheses, higher-usage households churn significantly more, reflecting price sensitivity rather than switching-cost lock-in.
- **Short customer tenure** (< 36 months) and younger age.
- **Poor home energy labels** (E, F, G) and older homes.
- **Online acquisition channel** and absence of email on file (weaker but still significant signals).

**Actionable Retention Strategies**

- Offer contract upgrades with attractive fixed-term discounts or bonuses to convert flexible-contract customers into locked-in contracts.
- Provide personalized energy-saving recommendations and bill-smoothing plans for high-usage households to reduce perceived cost burden.
- Implement onboarding loyalty programs and proactive welcome calls for newly acquired online customers.
- Target poorly insulated homes with subsidized insulation or green-energy bundle offers that simultaneously lower bills and improve energy labels.

### 7.4 Limitations

The study has three main limitations. First, the dataset is cross-sectional and lacks dynamic variables (e.g., recent bills, payment behavior, complaints, competitor offers), which are strong churn drivers in practice. Second, the artificial 50% churn rate inflates performance metrics and may cause overconfident probabilities in real deployment (actual Dutch churn ≈10–15% annually); recalibration will be required. Third, important external factors (energy price

volatility, regulatory changes, seasonal switching waves, and competitive intensity) are not captured, meaning model performance may degrade if market conditions shift significantly. Despite these constraints, the selected SVM remains the most accurate and actionable model given the available data, and regular retraining with enriched transactional and market data is recommended.

## VIII. AI Usage Statement

Artificial Intelligence (AI) was employed in this project as a supportive resource to facilitate both technical and communicative aspects of the analysis. Its primary role was in error correction, where AI assisted in identifying and resolving coding issues, thereby ensuring the smooth execution of statistical procedures and model implementation.

Beyond technical troubleshooting, AI contributed to conceptual understanding by clarifying complex analytical methods and statistical principles encountered during the study. This enhanced the reliability of the modeling process and supported a deeper comprehension of methodological choices.

AI was also utilized in the preparation of the written report. Specifically, it aided in improving grammar, structure, and clarity, ensuring that the documentation adhered to academic standards of precision and readability. By supporting both technical accuracy and effective communication, AI usage strengthened the overall quality and coherence of the project.

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